

CSE 400: Fundamentals of Probability in Computing21-22 Notes (From Chebyshev's Inequality)

Scribe Refactoring

1 Chebyshev's Inequality

Definition: For any random variable X with mean μ and variance σ^2 , $[\Pr(|X - \mu| \geq k\sigma) \leq \frac{1}{k^2}, \quad k > 0]$

Equivalent Form: $[\Pr(|X - \mu| \geq t) \leq \frac{\sigma^2}{t^2}]$

1.1 Proof

Let $Y = (X - \mu)^2$. Using Markov's inequality: $[\Pr((X - \mu)^2 \geq t^2) \leq \frac{\mathbb{E}[(X - \mu)^2]}{t^2} = \frac{\sigma^2}{t^2}]$

2 Interpretation

Chebyshev's inequality provides a universal bound that applies to *any* distribution with finite variance.

3 Example

If $\sigma = 2$, then: $[\Pr(|X - \mu| \geq 4) \leq \frac{1}{4}]$

4 Chernoff Bound

Motivation: Provides exponentially tighter bounds than Chebyshev.

Setup: Let $X = \sum X_i$ where X_i are independent Bernoulli.

Upper Tail: $[\Pr(X \geq (1 + \delta)\mu) \leq e^{-\frac{\mu\delta^2}{3}}, \quad 0 < \delta < 1]$

Lower Tail: $[\Pr(X \leq (1 - \delta)\mu) \leq e^{-\frac{\mu\delta^2}{2}}]$

5 Proof Sketch

Uses moment generating function (MGF) and Markov's inequality: $[\Pr(X \geq a) = \Pr(e^{tX} \geq e^{ta}) \leq \frac{\mathbb{E}[e^{tX}]}{e^{ta}}]$

Optimize over t .

6 Poisson Tail Bounds

For $X \sim \text{Poisson}(\lambda)$: $[\Pr(X \geq (1 + \delta)\lambda) \leq \left(\frac{e^\delta}{(1 + \delta)^{1 + \delta}}\right)^\lambda]$

7 Jensen's Inequality

Definition: If f is convex, then: $[f(\mathbb{E}[X]) \leq \mathbb{E}[f(X)]]$

7.1 Convex Function

f is convex if: $[f(\theta x + (1 - \theta)y) \leq \theta f(x) + (1 - \theta)f(y)]$

8 Applications

- Bounding tail probabilities
- Performance guarantees in randomized algorithms
- System reliability analysis

9 Case Study: Server Load

Tail bounds help estimate worst-case demand in systems.

10 Summary for Exam

- Chebyshev: Works for all distributions
- Chernoff: Strong exponential bounds
- Jensen: Links convexity and expectation
- Tail bounds: Essential for worst-case guarantees